Part

(a) $pr[J_k] = 1/k$ (2 marks)

~~X~~ Random varible (define properly)

$$x \to \begin{cases} 1 & p \to 1/k \\ 2 & \left(1-\frac{1}{k}\right) 1/k \\ \vdots & \\ i & \left(1-\frac{1}{k}\right)^{i-1} \frac{1}{k} \end{cases}$$

1 marks

$$E[X] = 1 \cdot \frac{1}{k} + 2 \cdot \left(1-\frac{1}{k}\right) 1/k + \cdots + i\left(1-\frac{1}{k}\right)^{i-1} \frac{1}{k}$$

— (1) (eq^n 1)

$$\left(1-\frac{1}{k}\right) E[X] = \left(1-\frac{1}{k}\right)\frac{1}{k} + 2\left(1-\frac{1}{k}\right)^2 \frac{1}{k} \cdots$$

— (2) eq^n 2

eq^n (1) − eq^n (2)

$$\frac{1}{k} E[X] = \left(1-\frac{1}{k}\right)^0 \frac{1}{k} + \left(1-\frac{1}{k}\right)\frac{1}{k} \cdots$$

sum of G.P.

$$\sum_{r=0}^{\infty} \left(1-\frac{1}{k}\right)^r = K$$

(1) marks

part (b) (3 marks)

define the Random varible

$$X_1 = \begin{cases} 1 \end{cases}$$

$$X_2 = \begin{cases} 1 & (1-1/k) \\ 2 & \left(\frac{1}{k}\right)(1-1/k) \\ \vdots \end{cases}$$

same for all — (1) marks

find the expectation

$$E[X_1] = 1$$

$$E[X_2] = \frac{k}{k-1}$$

$$E[X_3] = \frac{k}{k-2}$$

$$E\left[\sum_{i=1}^{k} X_i\right] = \sum_{i=1}^{k} \frac{k}{k-i+1}$$

$$= k \sum_{i=1}^{k} \frac{1}{k-i+1}$$

— (1) marks

$$k\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k}\right)$$

$$= k \log k$$

— (1) marks.

Part (c) (7 marks)

find $P_1 = \frac{1}{4}$ $P_2 = 1/2$ $P_3 = \frac{1}{4}$

$P_{12} = P_1 + P_2 = 3/4$

$P_{13} = P_1 + P_3 = 1/2$

$P_{23} = P_1 + P_3 = 3/4$

$P_{123} = P_1 + P_2 + P_3 = 1$

3 marks

Formula for ~~three~~ inclusion and exclusion of three events

$$\Rightarrow \left(\frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3}\right) - \left(\frac{1}{P_{12}} + \frac{1}{P_{13}} + \frac{1}{P_{23}}\right)$$

2 marks

put all values.

$$(4+2+4) - \left(\frac{4}{3} + 2 + \frac{4}{3}\right) + 1$$

$$11 - \left(\frac{4+6+4}{3}\right)$$

$$11 - \frac{14}{3} = \frac{33-14}{3} = \frac{19}{3} = 6.33$$

2 marks